

AVANISH GADHIKAR

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EDUCATION

CARNEGIE MELLON UNIVERSITY, TEPPER SCHOOL OF BUSINESS

Pittsburgh, PA

Master of Science in Computational Finance – MSCF

GRE Quant: 170/170

12/27

- Courses in financial data science, options, investments, fixed income, stochastic calculus

INDIAN INSTITUTE OF TECHNOLOGY, GUWAHATI

Assam, India

Bachelor of Technology in Mathematics and Computing

CGPA: 9.01/10.0 (Ranked 4th of 79)

06/26

- **Minor** in Data Science and Artificial Intelligence (DS & AI)
- Key courses included linear algebra, calculus, option pricing, machine learning, probability & statistics
- **TA** for a course 'Financial Valuation and Portfolio Analytics'; part of a study group on 'DeFi and FinTech'
- **Programming skills:** Python, C, C++, MATLAB

RESEARCH/PROJECTS

ROTMAN INTERACTIVE TRADER STRATEGY ENGINE (MSCF Trading Competition)

09/26

- **Algorithmic Trading:** Built a Python DMA trading system for ETF arbitrage, volatility arbitrage, tender evaluation, hedging, and emergency flattening across ETF, FX, equity, and options instruments.
- **Depth-Aware Arbitrage:** Designed FX-adjusted ETF/basket pricing logic using 20-level order book VWAP; exploiting mispricing through 10,000-share child orders under 300K gross / 200K net risk limits.

MODELLING OF MICROSTRUCTURES OF ELECTRONIC MARKETS

08/25 – 04/26

- **Poissonian Order Flows:** Simulated event-driven LOB dynamics using independent Poisson flows for market, limit orders, and cancellations; calibrated MSFT L10 LOBSTER data with lambda of 0.714.
- **Hawkes Calibration:** Modeled 6 self-exciting order-flow streams using exponential decay kernels with beta set to 1.5; estimated baseline intensities and excitation matrix via MLE and L-BFGS-B optimization.
- **GAN Architecture:** Implemented a conditional WGAN-GP to generate centered LOB snapshots.

MONTE CARLO SIMULATIONS

08/24 – 11/24

- **Mathematical Simulations:** Built simulations for random sampling and estimation using Inverse Transform, Acceptance-Rejection, and Box-Muller methods; achieved variance reduction through Antithetic Variables, Control Variates, and Stratification.
- **Optimizations & Pricing:** Modeled Geometric Brownian Motion and Jump Diffusion processes to price European, American and Asian options; utilizing Sobol, and Halton sequences for faster convergence.

xv6 OPERATING SYSTEM

08/24 – 11/24

- **Process Scheduling:** Extended the xv6 kernel to include three custom schedulers (Lottery, SJF, and Hybrid); implemented lazy allocation with an LRU-based swapping to optimize kernel memory.
- **File Storage:** Evaluated system Input/Output performance across two file systems (ZFS and EXT4), profiling write throughput, deduplication efficiency, and CPU overhead via system benchmarking tools.

EXPERIENCE

SALESFORCE

Bengaluru, India

Associate Member of Technical Staff, Intern

05/25 – 08/25

- **Agentic Systems:** Engineered Python AIOps package comprising 7 diagnostic tools within the Warden framework, automating queries to reduce on-call Time To Diagnose from hours to 2–4 minutes.
- **API Integration:** Architected custom Slack-integrated applications, to proactively track channel metrics, alert relevant personnel, and accelerate the resolution of outbound access and 403, 503 type errors.
- **Production Reliability:** Collaborated across infrastructure teams to ensure compatibility with workflows and high-availability environments; gaining experience in secure cloud architectures and live operations.

ADDITIONAL INFORMATION

- **Interests:** Football (Man Utd), Running (10k PB 55mins), F1 (Super Max), Music, Poker
- **Leadership:** Technical Secretary of Barak Hostel, IITG (Led a contingent of 100+ people to victory in an inter-hostel technical competition); Captained the high-school football team.
- **Languages:** English, Hindi, Marathi (Native)